

1. **Combustion:** fuel (hydrocarbon) + oxygen **V** carbon dioxide + water

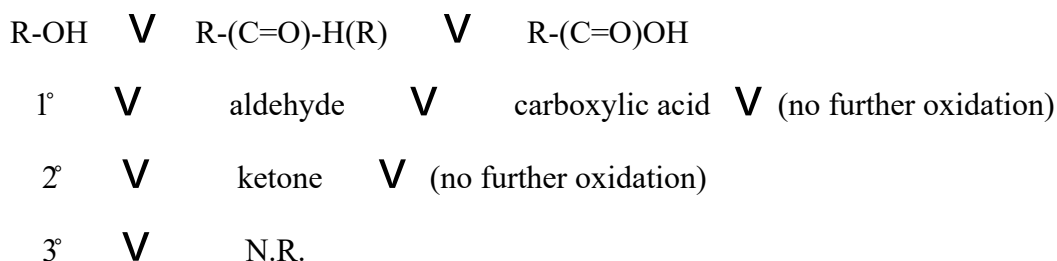
ex. $\text{C}_5\text{H}_{12} + 8 \text{O}_2 \text{ V } 5 \text{CO}_2 + 6 \text{H}_2\text{O}$ (this is the only reaction we balance)

2. **Addition:** alkene + X-Y **V** X-Y add across the double bond (X-Y = H_2 , Br_2 , HBr , or HOH to make an alcohol)

ex. $\text{CH}_3\text{CH}=\text{CHCH}_3 + \text{HBr} \text{ V } \text{CH}_3\text{CH}-\text{CHCH}_3$ (also: Markovnikov's Rule!)
 $\begin{array}{cc} | & | \\ \text{H} & \text{Br} \end{array}$ (H goes to C with most H's already)

3. **Dehydration** of Alcohol: alcohol plus acid and heat (180°C) to give alkene plus water (opposite of addition of HOH to alkene above) **OR** heat (140°C) to give the ether.

4. **Oxidation:** [O] addition of oxygen atom(s) and/or loss of H atom(s)



5. **Benedict's (blue copper to red solid) and Tollens' (formation of silver mirror) Tests**

aldehydes (+) ketones (–) monosaccharides (+) disaccharides (all +, except sucrose –)

6. **Hydrolysis:** (or reverse hydrolysis) look for places to split out (or add) water

e.g. formation of **glycosidic linkage**, or a **peptide bond** or a **phosphodiester linkage**

7. **Esterification:** alcohol + carboxylic acid yields ester + water

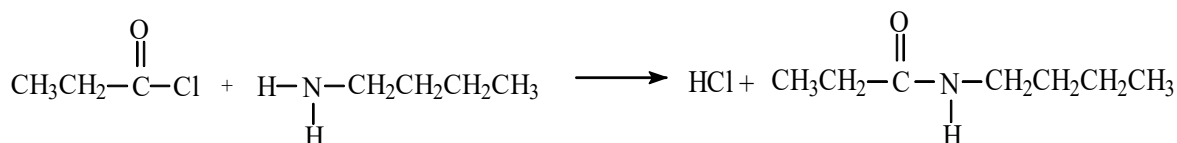
ex. $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{C(=O)-OH} \text{ V } \text{CH}_3\text{CH}_2\text{CH}_2\text{C(=O)-O-CH}_2\text{CH}_3 + \text{H}_2\text{O}$

naming of esters requires recognizing the alcohol fragment and the acid fragment; name is alkyl (for the alcohol) alkanoate (for the acid); this ester is *ethyl butanoate*

8. **Saponification:** fatty acid + NaOH (aqueous) to give carboxylic acid salt (soap)

$\text{CH}_3(\text{CH}_2)_{12}\text{-C(=O)-O-H} + \text{NaOH} + \text{HOH} \text{ V } \text{CH}_3(\text{CH}_2)_{12}\text{-C(=O)-O}^-\text{Na}^+$

9. **Amide Formation** - Amine plus acid chloride yields amide plus HCl



10. **Phosphodiester** – formation of the 5' and 3' O to phosphate linkage in DNA, RNA

Classes of Compounds. Be able to recognize, draw, and name them. NOMENCLATURE!!

alkanes	alcohols	ketones	esters	cycloalkanes
alkenes	ethers	aromatics	carboxylic acids	cycloalkenes
alkynes	aldehydes	amines	amides	

Know their physical properties too! Which ones can hydrogen bond? Which are high m.p./b.p.? Which are expected to be water soluble?

Be able to recognize. Know properties.

fats (lipids)	proteins	carbohydrates
enzymes	DNA	RNA

Formulas and Isomers

Know how the formula of a compound can tell you what are the possible classes of the compound. How does the number of hydrogens present help you know if a $C = C$, or $C \equiv C$, or ring is present? Be able to draw isomers of a given formula. *Cis*- and *trans*- are special isomer terms.